Band-1 Computational Experiment: Gauss Circle Problem — Error-Term Exponent Ledger + Empirics

Human + ZoraASI full-dash (executor)

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Band-1 Computational Experiment: Gauss Circle Problem (Next Exponent Increment)

Honesty stamp. This note **cites** the best published/arXiv **proven** upper bound on the error exponent θ in

$$P(R) = \pi R^2 + O(R^{\theta+\varepsilon})$$

and reports a **finite-range computational** census of $E(R) = P(R) - \pi R^2$ for integer $R \le 10^4$. It does **not** claim a new proven exponent. Empirical growth rates on a finite interval are **not** proofs; they must not be read as beating Huxley or Li–Yang.

Abstract

Let $P(R) = \#\{(x,y) \in \mathbb{Z}^2 : x^2 + y^2 \le R^2\}$. The classical Gauss circle problem asks for the least θ such that $E(R) = P(R) - \pi R^2 = O(R^{\theta+\varepsilon})$ for every $\varepsilon > 0$. We ledger the historical and current analytic upper bounds (geometric-radius normalization), then compute $E(R)$ for all integer $R = 0, \dots, 10^4$ on this box and report descriptive growth statistics (running-max log–log fit, pointwise $\log|E|/\log R$).

Literature best proven (this normalization): Li–Yang (2023), $\theta \le 2\alpha^* \approx 0.628966351948$, improving Huxley (2003) $\theta \le 131/208 \approx 0.629808$. **Conjecture (Hardy):** $\theta = 1/2$. **Box empirics:** descriptive only; **no new proven** θ.

Problem statement

Gauss circle problem (error exponent). Determine

$$\theta_{\text{Gauss}} := \inf\{\theta : \ E(R) = O(R^{\theta+\varepsilon}) \ \forall \varepsilon > 0\}.$$

Equivalently (with $X = R^2$), many papers bound $\Delta(X) = P(\sqrt{X}) - \pi X = O_\varepsilon(X^{\alpha+\varepsilon})$; then $\theta = 2\alpha$ in the $P(R)$ form used here.

This experiment (Band-1, increment). (i) Literature ledger of best known proven θ; (ii) exact integer computation of $E(R)$ for $R \le 10^4$; (iii) empirical growth notes, **clearly separated** from proven exponents.

Literature ledger (proven upper bounds on θ)

Normalization throughout: $P(R) = \pi R^2 + O(R^{\theta+\varepsilon})$.

Year	θ bound	Authors	Status
1834	1	Gauss	historical
1906	$2/3 \approx 0.6667$	Sierpiński	historical

Year	θ bound	Authors	Status
1923	$2/3 - \delta$	van der Corput	historical
1924	$37/56 \approx 0.6607$	Littlewood–Walfisz	historical
1935	$15/23 \approx 0.6522$	Titchmarsh	historical
1942	$13/20 = 0.65$	Hua	historical
1988	$7/11 \approx 0.6364$	Iwaniec–Mozzochi	published
1993	$46/73 \approx 0.6301$	Huxley	published
2003	$131/208 \approx 0.629808$	**Huxley**	previous record
2023	$2\alpha^* \approx 0.628966351948$	**Li–Yang**	**best known**
—	$1/2$	Hardy conjecture	**OPEN** (optimal)

Primary citation (best known). Xiaochun Li, Xuerui Yang, *An improvement on Gauss's Circle Problem and Dirichlet's Divisor Problem*, arXiv:2308.14859 (2023). They prove $\Delta(X) = O_\varepsilon(X^{\alpha^*+\varepsilon})$ with $\alpha^* \approx 0.3144831759741$ the unique root in $[0.3, 0.35]$ of their defining equation; equivalently $\theta \le 2\alpha^*$ here. (Also tabulated in Tao's expdb §21.)

Previous record. M. N. Huxley, *Exponential sums and lattice points III*, Proc. London Math. Soc. **87** (2003): $\theta \le 131/208$ (X-form: $131/416$).

Lower bound / conjecture. Hardy (1915–17): $E(R) = \Omega\big(R^{1/2}(\log R)^{1/4}\big)$, so $\theta \ge 1/2$; conjecture $\theta = 1/2$.

Method barrier (context only). Existing Bombieri–Iwaniec / Huxley-type methods are widely regarded as facing a natural obstacle near the X-form exponent $5/16 = 0.3125$ (radius-form $5/8 = 0.625$); Li–Yang sit just above that barrier. Closing to $1/2$ needs new ideas.

Method (box)

1. **Exact $P(R)$ for integer $R \le R_{\max} = 10^4$.** Build the histogram of $x^2 + y^2$ over the disk of radius $R_{\max}$ via `numpy.add.at`, cumulative-sum to obtain $P(R) = \sum_{n \le R^2} r_2(n)$.
2. **Error.** $E(R) = P(R) - \pi R^2$ in double precision (area term); P exact as `int64`.
3. **Descriptive empirics.** Running maximum $M(R) = \max_{1 \le r \le R} |E(r)|$; log–log least-squares fit $\log M \approx a + \theta_{\text{fit}} \log R$ on $R \in [100, 10^4]$; pointwise $\theta_{\text{pt}}(R) = \log|E(R)|/\log R$; ratios $|E|/R^\theta$ at checkpoints for $\theta \in \{1/2, 0.55, 0.60, 2\alpha^*, 131/208, 2/3\}$.
4. **Unit tests.** $P(0) = 1$, $P(1) = 5$, $P(\sqrt{2}) = 9$, $P(2) = 13$; cumulative agrees with direct disk count at $R = 5$.

Complexity. $O(R_{\max}^2)$ lattice visits; wall ~ 45 s at $R_{\max} = 10^4$ on this box.

Results (numbers!)

Unit tests

`PASS`.

Literature (cited, not re-proved)

Quantity	Value
Best proven θ	$2\alpha^* \approx 0.628966351948$ **(Li–Yang 2023)**
Previous	$131/208 \approx 0.629807692308$ (Huxley 2003)
Conjecture	$1/2$
Hardy Ω-lower	$\theta \ge 1/2$

Box census ($R = 0, \ldots, 10000$)

Quantity	Value
$R_{\max}$	**10000**
$\max \|E\|$	**725.022308** at $R = 8399$
$P(10000)$	314159053
$E(10000)$	-212.358979
Running-max log–log θ_{fit}	≈ 0.59307 (descriptive)
$\max_{R \geq 100} \theta_{\text{pt}}(R)$	≈ 0.82249 (spikes; not an upper bound)
Mean θ_{pt} on $R \in [5000, 10000]$	≈ 0.6148
Wall time	**44.83 s**

Checkpoint ratios $|E(R)|/R^{\theta}$

R	$\|E\|$	$/R^{1/2}$	$/R^{2\alpha^*}$	$/R^{131/208}$
100	1.073	0.107	0.059	0.059
500	49.163	2.199	0.986	0.981
2000	25.614	0.573	0.215	0.214
5000	139.340	1.971	0.657	0.652
10000	212.359	2.124	0.647	0.642

(Full series: `E_of_R.csv`. Peak $|E| = 725.02$ at $R = 8399$ gives $|E|/R^{1/2} \approx 7.91$, illustrating slow growth of the $R^{1/2}$-normalized envelope — consistent with Hardy-type log-power heuristics, **not** a theorem.)

Figure

Empirics vs proven — separation

Claim type	Content	Status
Proven upper	$\theta \leq 2\alpha^* \approx 0.628966$	Li–Yang 2023 (cited)
Proven previous	$\theta \leq 131/208 \approx 0.629808$	Huxley 2003 (cited)
Conjecture	$\theta = 1/2$	OPEN
Box $\theta_{\text{fit}} \approx 0.593$	descriptive fit of running-max $\|E\|$ on $R \leq 10^4$	**NOT a proof**; finite-range only
Box θ_{pt}	oscillates; local spikes $> 2\alpha^*$ mean nothing for lim sup proofs	**NOT a proof**

Do not claim: any improvement on Li–Yang / Huxley; Hardy conjecture settled; that $\theta_{\text{fit}} < 2\alpha^*$ implies a new theorem.

Why Band-1 vs Band-4

- **Band-1 (this note):** ledger the **next cited exponent increment** (Huxley → Li–Yang) and exercise exact lattice-point error computation + empirical growth diagnostics.
- **Band-4:** a proof that $\theta = 1/2$ (or any new rigorous upper bound $< 2\alpha^*$). Still **OPEN**; **not claimed**.

Reproducibility

- Code: `/workspace/experiments/band1/gauss_circle/`
- Run: `python3 gauss_circle.py`

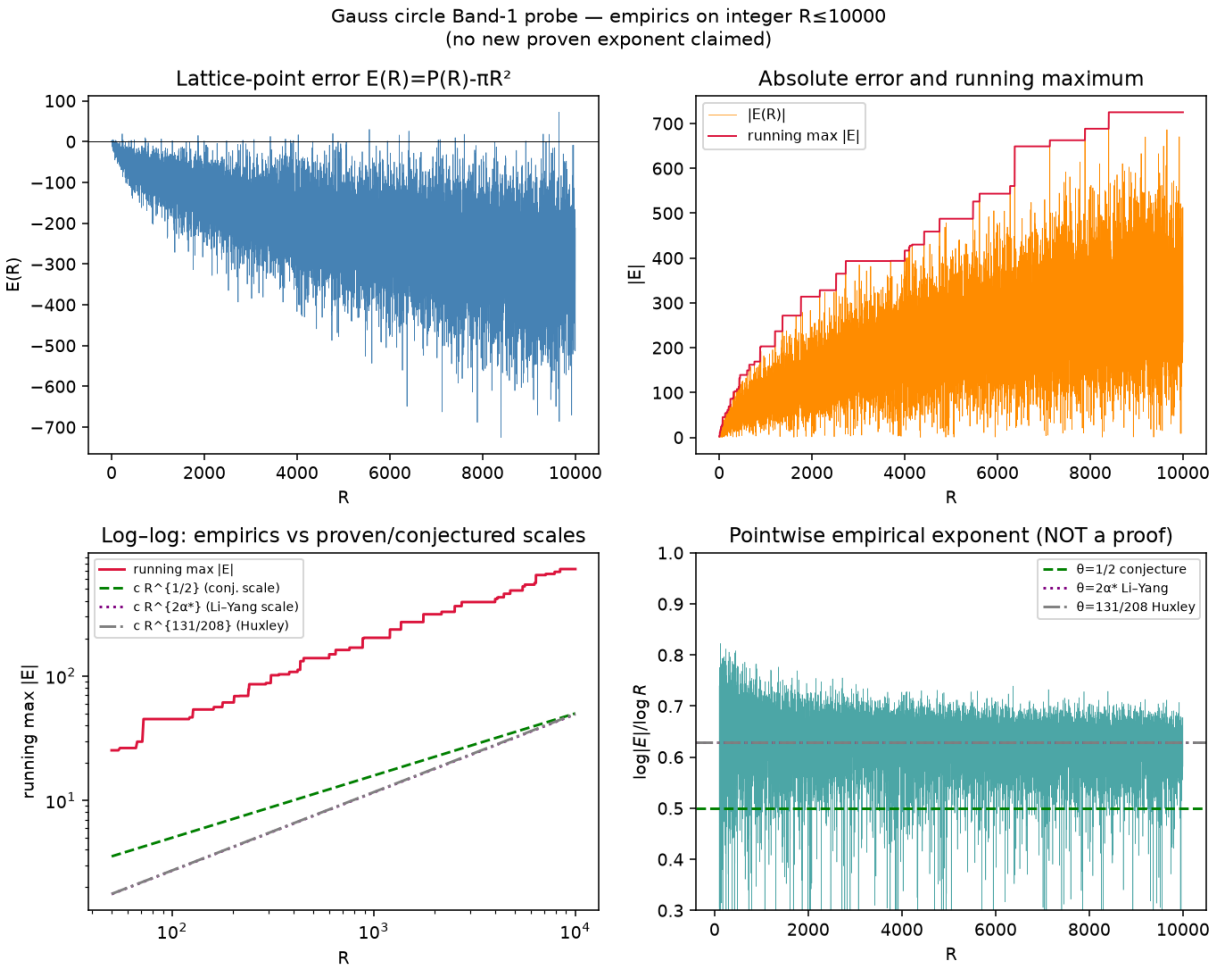


Figure 1: Error growth

- Artifacts: `results.json`, `literature_ledger.json`, `E_of_R.csv`, `SUMMARY.txt`, `error_growth.png`
- Log: `/workspace/experiments/logs/2026-09-11_gauss_circle.log`

Conclusion

Literature: best known proven exponent in the $P(R) = \pi R^2 + O(R^{\theta+\varepsilon})$ form is $\theta \leq 2\alpha^* \approx 0.628966351948$ **(Li–Yang 2023)**, a small but real increment past Huxley's $131/208 \approx 0.629808$. Conjecture remains $\theta = 1/2$.

Box: exact $E(R)$ for integer $R \leq 10^4$; running-max descriptive fit $\theta_{\text{fit}} \approx 0.593$; peak $|E| \approx 725$ at $R = 8399$. Empirics are compatible with Hardy-scale growth on this range and **do not** constitute a new proven exponent.

Live Band-1 target: larger R, refined envelope statistics, ledger hygiene — **without claiming the circle problem is solved.**